Review Test Submission: Section 11 Model Diagnostics

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Course	2021-ST311 Applied Statistics I
Test	Section 11 Model Diagnostics
Started	04/12/20 15:58
Submitted	04/12/20 16:03
Due Date	09/12/20 12:00
Status	Completed
Attempt Score	21 out of 21 points
Time Elapsed	5 minutes
Instructions	
Results Displayed	Submitted Answers, Incorrectly Answered Questions

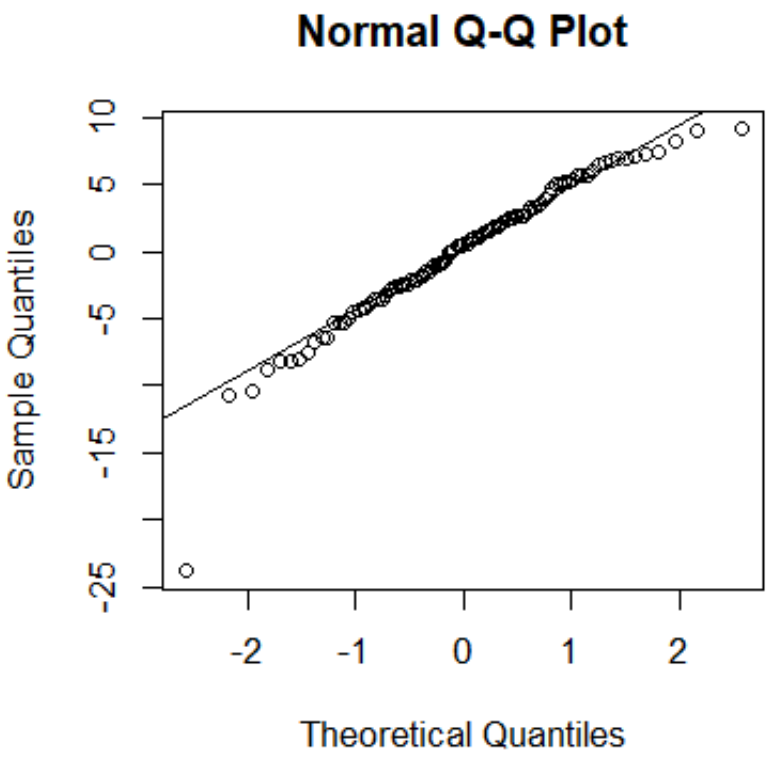
Question 1

3 out of 3 points

A simple linear model, $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, is fitted to sample data observed for n = 100 individuals.

The output below summarises the fit:

```
Coefficients: Estimate Std. Error t value Pr(>|t|)
(Intercept)   21.04    1.0420     20.1931  0.0000
    X          1.87    0.0911     20.5142  0.0000
---
Residual standard error: 5.195 on 98 degrees of freedom
```



An individual in the sample was observed to have predictor value X = 9 and observed response Y = 14.

Find the value of the standardised residual for this individual.

Round to 2 dec places.

Selected Answer: -4.6

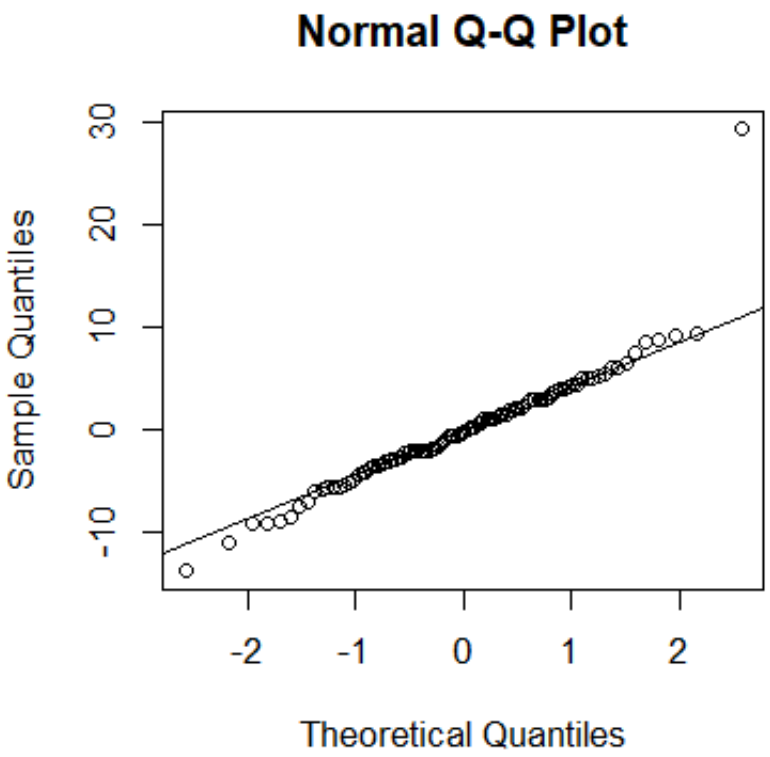
Question 2

3 out of 3 points

A simple linear model, $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ Y_i=\beta_0+\beta_1X_i+\epsilon_i , is fitted to sample data observed for n = 100 individuals.

The output below summarises the fit:

```
Coefficients: Estimate Std. Error t value Pr(>|t|)
(Intercept)     1.40     1.2155     1.1548  0.2510
      X         1.09     0.0989    11.0306  0.0000
---
Residual standard error: 5.4279 on 98 degrees of freedom
```



An individual in the sample was observed to have predictor value X = 14 and observed response Y = 46.

Find the value of the standardised residual for this individual.

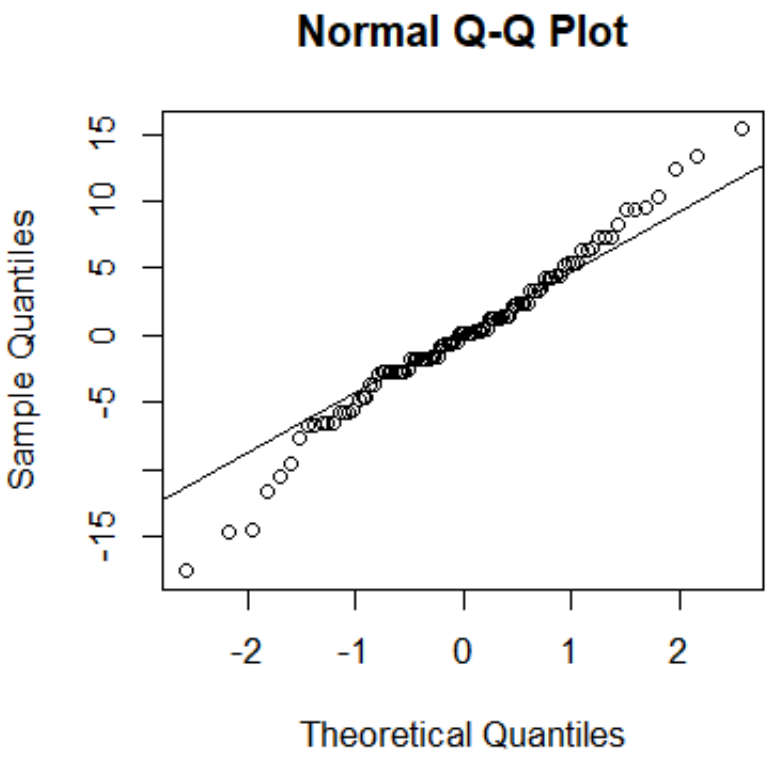
Round to 2 dec places.

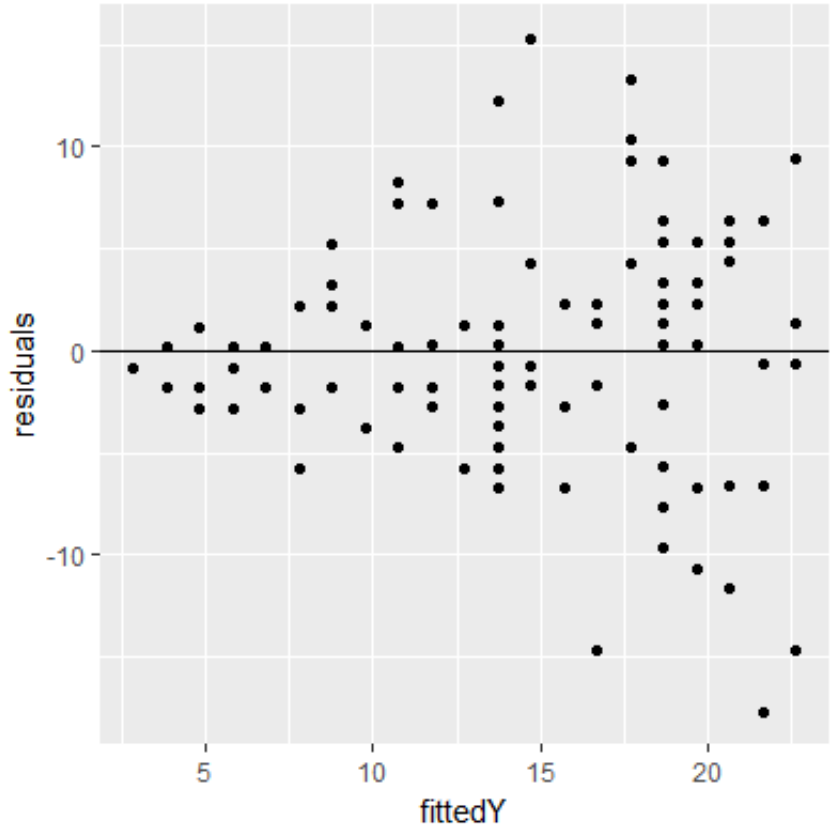
Selected Answer: 5.41

Question 3

5 out of 5 points

A linear regression model is fitted to data observed for a sample of individuals. The output below provides diagnostics of the fitted residuals.






```
Anderson-Darling normality test

data: fit$residuals
A = 0.68712, p-value = 0.07062
```

Indicate which of the following statements are true:

Selected Answers: c. The anderson-darling test indicates no evidence the normality of residuals assumption is violated.

e. The qq-plot suggests the normality of residuals assumption is not violated since points lie along the diagonal line of equality.

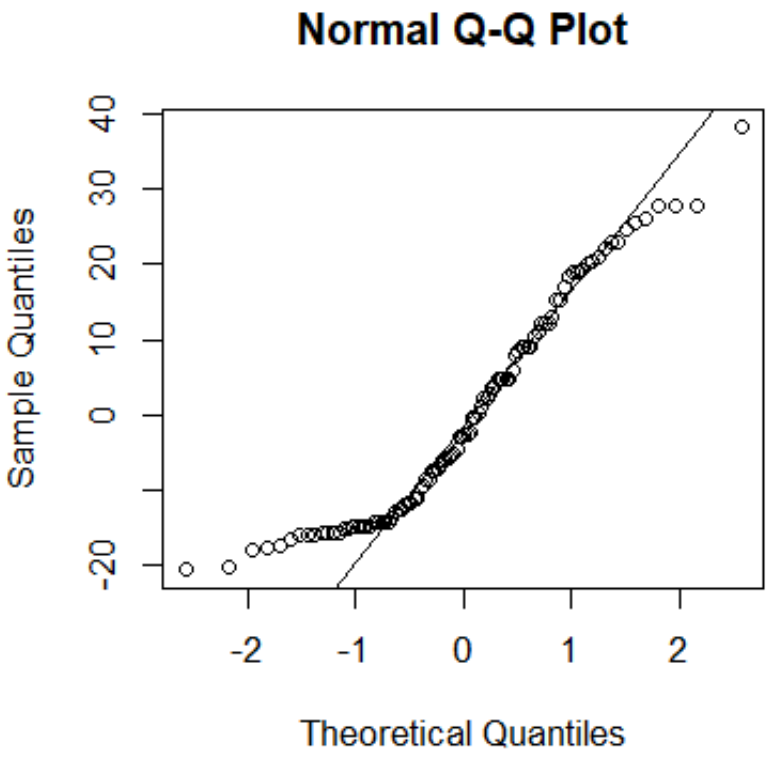
f. The residual versus fits plot indicates non-constant variance assumption is violated.

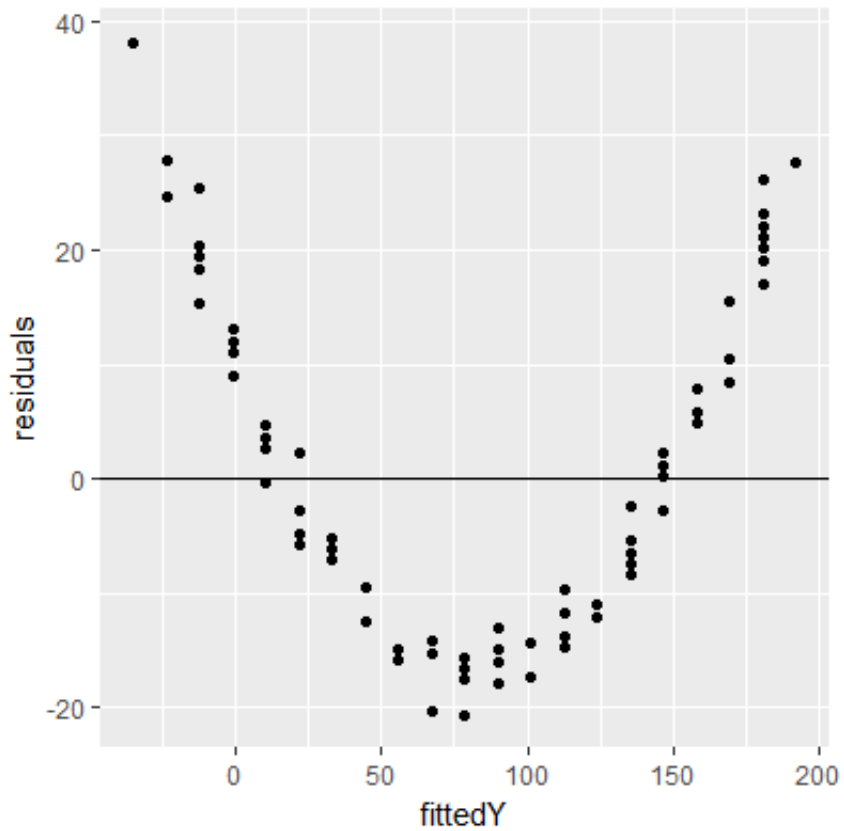
h. Consider a log transformation of the response variable.

Question 4

5 out of 5 points

A linear regression model is fitted to data observed for a sample of individuals. The output below provides diagnostics of the fitted residuals.






```
Anderson-Darling normality test

data: fit$residuals
A = 2.3564, p-value = 5.246e-06
```

Indicate which of the following statements are true:

Selected Answers: b. The anderson-darling test indicates the normality of residuals assumption is violated.

d. The qq-plot suggests the normality of residuals assumption is violated since there is strong deviation from the diagonal line of equality.

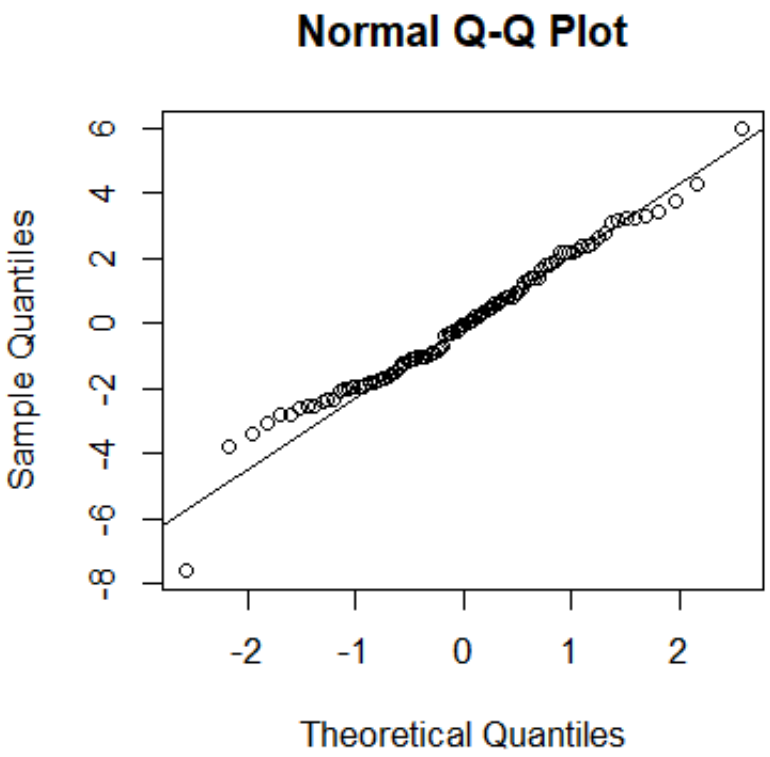
g. The residual versus fits plot indicates non-constant variance assumption is not violated.

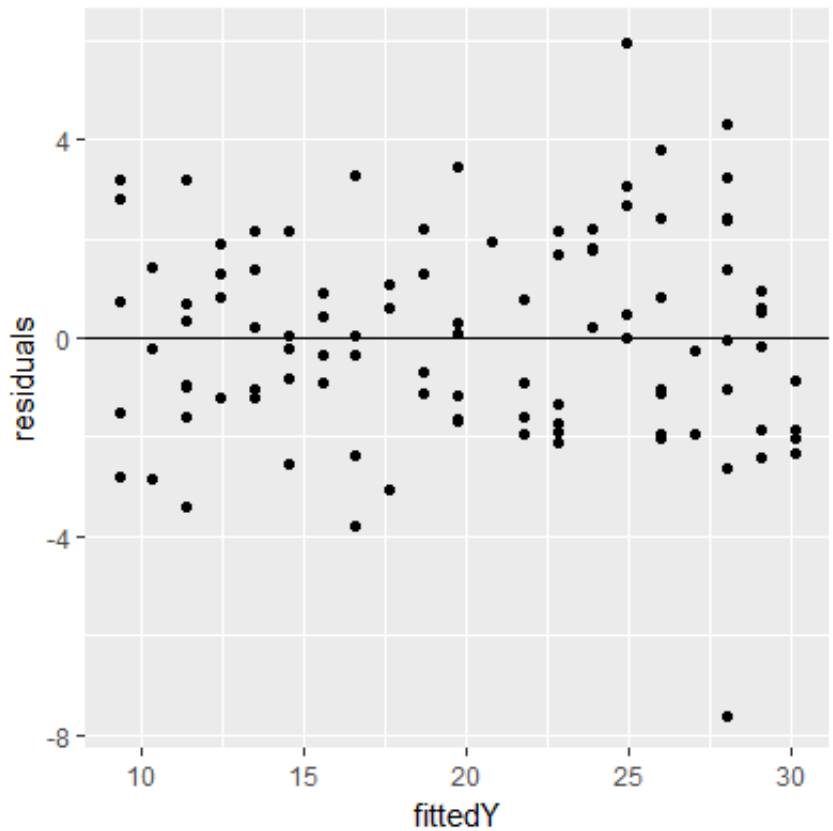
i. Consider a quadratic or higher order term in the model.

Question 5

5 out of 5 points

A linear regression model is fitted to data observed for a sample of individuals. The output below provides diagnostics of the fitted residuals.






```
Anderson-Darling normality test

data: fit$residuals
A = 0.43802, p-value = 0.2894
```

Indicate which of the following statements are true:

Selected Answers: c. The anderson-darling test indicates no evidence the normality of residuals assumption is violated.

e. The qq-plot suggests the normality of residuals assumption is not violated since points lie along the diagonal line of equality.

g. The residual versus fits plot indicates non-constant variance assumption is not violated.

j. All assumptions of the linear regression model are satisfied.

Monday, 4 January 2021 14:16:27 o'clock GMT

← OK